MASTER COPY

BIOMEDICAL LIMITED

ANNUAL PRODUCT QUALITY REVIEW

(YEAR – 2025)

Protocol No.	**: BML/QA/APQR/QA/002**
Protocol Date	: 23/04/2024
Total No. of Pages	**:**
PRODUCT NAME	**:** Bioflex Dextrose 5%w/v Intravenous Injection B.P 500ml
GENERIC NAME	**:** Dextrose 5% w/v Intravenous Injection B.P 500ml
COMPOSITION	: Each 100ml contains Dextrose 5g Water for injection B.P q.s
PACK SIZE	: 500ml in a printed PVC pouch
PACKAGING TYPES AVAILABLE	**:** 500ml Polyvinylchloride PVC pouch Corrugated cardboard boxes carton
MASTER FORMULAR NUMBER	: BML/QA/MFR/003
SHELF LIFE:	3 Years
TIME PERIOD COVERED	: JANUARY – DECEMBER, 2025
PRODUCT CODE	: D

BIOMEDICAL LIMITED

1, Ohimege Road, Industrial Estate, Gaa- Imam, Ilorin Kwara State Nigeria.

CONTROL COPY

Sr. No	Subject	Name	Designation	Sign/Date
1	Compiled by	Olawale Ogunleye	Quality Assurance Officer	
2	Reviewed by	Robiu Idowu M.	Production Pharmacist	
3	Approved by	Koroma Abigael A.	Quality Assurance Manager	
4	Authorized by	Agboola L. A.	Factory Manager	

1.0 SCOPE:

This covers internal annual products quality review, and all these parameters are monitored; changes in physical appearance of the product, chemical analysis (pH, Assay), microbiological analysis (Sterility and Pyrogen), and post market surveillance.

2.0 DESCRIPTION OF PRODUCT

Bioflex Dextrose 5%w/v Intravenous Injection is a clear colorless sterile solution in a PVC pouch with double port for easy administration. Formulated for the treatment of insulin hypoglycemia to restore glucose levels, the hypertonic solution is carefully controlled to ensure the glucose level of 5g/100ml. Careful control of all other parameters as mentioned in the scope and under the section for the review of the quality control of the product in this piece are assured at all times.

3.0 REVIEWS

3.1 REVIEW OF PRODUCT BATCHES MANUFACTURED

A batch of Bioflex Dextrose 5%w/v Intravenous Injection comprises of about 3,050 bags. Total No. of Batches produced within the period January 2025 to December 2025 is 286 batches. I.e. from Batch no.: D25001 to D25286; all the batches produced have shown reliability of the manufacturing and the testing procedures in delivering constant and reproducible good product quality.

3.2 REVIEW OF STARTING AND PRIMARY PACKAGING MATERIALS USED IN THE FPP

The starting materials for the manufacture of Bioflex Dextrose 5%w/v Intravenous Injection are Dextrose monohydrate (active ingredient) and Water for injection (excipient). Dextrose monohydrate is sourced from approved vendors while water for injection is produced in house. The starting materials are each tested to comply with respective pharmacopeia and in-house specifications. The table below shows the different consignment of dextrose monohydrate received and used in the manufacture of Bioflex Dextrose 5%w/v Intravenous Injection within the year under review. All the batches of the active material received met the specifications of pharmacopeia.

3.2.1 VENDORS

All the raw materials used in the production of these batches are sourced from the approved vendors only.

3.2.2 RAW MATERIAL (API) MANUFACTURER/ SUPPLIER PERFORMANCE REVIEW

Name of API	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignment / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
Dextrose Monohydrate	Weifang Shengtai Medicine Co. Ltd. China	Weifang Shengtai Medicine Co. Ltd. China	**7**	**7**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Bdv Behrens GmbH Hamburg, Germany.	**1**	**1**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Kuber, Fzco Dubai Silicon, Oasis.	**2**	**2**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Farmasino Co.Ltd China.	**2**	**2**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Bdv Behrens GmbH Hamburg, Germany.	**2**	**2**	**Nil**	**Nil**

3.2.3 PRIMARY PACKAGING MATERIAL

The FPP is contained in poly vinyl chloride (PVC) self-collapsible soft bag (pouch) bought as a finished product from the manufacturer of the PVC pouch. Manufacturer COA of the PVC is reviewed and testing done in-house to confirm the quality specifications stated by the manufacturers as conform to in-house specifications set for the primary packaging material.

Name of Primary packaging Material	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignments / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
PVC pouches	Jiangsu Medwell Medical	Jiangsu Medwell Medical	14	14	Nil	Nil

	Equipment Co. Ltd. China	Equipment Co. Ltd. China				
	Shanghai Kohope Medical Devices, Limited, China.	Shanghai Kohope Medical Devices, Limited, China.	2	2	Nil	Nil

3.3 REVIEW AND STATISTICAL ANALYSIS OF QUALITY CONTROL AND IN-PROCESS CONTROL

3.3.1 SPECIFICATIONS

Appearance: Clear, colorless and particles free solution

Dextrose B.P.: 95 – 105%

pH: 3.20 – 6.50

Hydroxymethylfurfural: NMT 0.25

3.3.2 IN-PROCESS RESULTS:

The summary of in-process results of all the batches manufactured was prepared and the table showing the results presented. The summary of in-process results for the chemical analysis and microbiological analysis of the batches manufactured is shown

Chemical Analysis of In-process product Result

S/N	BATCH NO	Dextrose Assay (95 – 105%)	pH (sample) (3.2 – 6.50)	pH (WFI) (5.0 – 7.0)
1	D25001	98.67	5.75	6.2
2	D25002	98.37	5.86	6.28
3	D25003	97.69	5.88	6.33
4	D25004	99.34	5.26	6.25
5	D25005	98.12	5.65	6.15
6	D25006	97.25	5.78	6.43
7	D25007	95.77	5.9	6.05
8	D25008	97.16	5.74	5.83
9	D25009	97.06	5.68	5.73
10	D25010	97.2	5.51	5.64
11	D25011	97.03	5.43	5.65
12	D25012	97.25	5.51	5.64
13	D25013	98.22	5.48	6.16
14	D25014	97.3	6.1	6.31
15	D25015	96.31	5.72	6.32

16	D25016	97.18	6.08	6.78
17	D25017	96.27	5.92	6.61
18	D25018	97.88	6.05	6.35
19	D25019	98.12	5.95	6.32
20	D25020	97.75	5.87	6.3
21	D25021	97.86	5.92	6.41
22	D25022	97.37	6.03	6.47
23	D25023	97.36	5.91	6.36
24	D25024	97.78	5.99	6.41
25	D25025	98.71	5.84	6.43
26	D25026	98.85	5.72	6.32
27	D25027	98.99	5.76	6.29
28	D25028	98.88	5.63	6.11
29	D25029	98.83	5.66	5.86
30	D25030	97.44	5.77	6.1
31	D25031	98.71	5.7	6.05
32	D25032	98.41	5.9	6.25
33	D25033	98.85	5.65	6.01
34	D25034	98.9	5.63	6.06
35	D25035	97.44	5.7	6.19
36	D25036	96.67	5.75	6.22
37	D25037	98.86	5.6	6.1
38	D25038	96.87	5.64	6.15
39	D25039	98.39	5.65	6.12
40	D25040	97.39	6.06	6.16
41	D25041	98.79	5.81	6.32
42	D25042	98	5.91	6.29
43	D25043	98.88	5.94	6.25
44	D25044	98.52	5.96	6.13
45	D25045	98.48	5.9	6.21
46	D25046	98.96	5.94	6.22
47	D25047	98.48	5.62	6.35
48	D25048	98.96	5.96	6.48
49	D25049	98.9	5.8	6.18
50	D25050	97.8	5.79	6.11
51	D25051	98.67	5.71	6.23
52	D25052	98.83	5.74	6.1
53	D25053	98.86	6.01	6.44
54	D25054	98.71	5.98	6.47
55	D25055	98.81	5.88	6.35
56	D25056	97.92	5.94	6.28
57	D25057	98.9	5.71	6.16
58	D25058	98.77	5.79	6.13
59	D25059	98.73	5.63	6.05
60	D25060	98.2	5.64	6.06
61	D25061	97.9	5.61	6.07

62	D25062	98.07	5.73	6.11
63	D25063	98.37	5.87	6.1
64	D25064	97.88	5.71	6.2
65	D25065	98.56	6.01	6.25
66	D25066	98.33	6.05	6.27
67	D25067	98.96	6.45	6.74
68	D25068	98.62	6.38	6.71
69	D25069	98.92	6.07	6.52
70	D25070	98.8	6.13	6.5
71	D25071	98.86	6.11	6.42
72	D25072	98.85	5.93	6.38
73	D25073	97.37	5.9	6.08
74	D25074	98.86	5.73	6.14
75	D25075	97.61	5.77	6.13
76	D25076	98.73	5.69	6.08
77	D25077	97.99	5.89	6.17
78	D25078	98.48	5.8	6.13
79	D25079	97.9	5.73	6.09
80	D25080	96.34	5.83	6.19
81	D25081	97.94	5.13	6.02
82	D25082	96.48	6.13	6.6
83	D25083	96.78	5.94	6.23
84	D25084	98.94	6.05	6.21
85	D25085	98.48	6	6.25
86	D25086	98.86	5.97	6.26
87	D25087	96.84	5.87	6.34
88	D25088	96.74	5.9	6.3
89	D25089	98.47	6.01	6.4
90	D25090	97.9	5.95	6.33
91	D25091	98.64	5.8	6.25
92	D25092	97.52	5.84	6.37
93	D25093	96.7	5.88	6.38
94	D25094	97.06	5.93	6.42
95	D25095	97.94	5.9	6.35
96	D25096	97.33	5.77	6.39
97	D25097	96.91	5.8	6.27
98	D25098	97.58	5.91	6.35
99	D25099	98.79	4.75	5.15
100	D25100	98.71	5.6	5.21
101	D25101	98.66	5.73	6.3
102	D25102	97.58	5.49	6.15
103	D25103	98.9	5.86	6.09
104	D25104	98.28	5.73	6.24
105	D25105	98.52	5.81	6.44
106	D25106	97.52	5.92	6.47
107	D25107	98.06	6.06	6.18

108	D25108	97.08	5.87	6.23
109	D25109	98.99	5.7	6.4
110	D25110	97.04	5.81	6.42
111	D25111	98.47	5.9	6.38
112	D25112	97.78	6.03	6.41
113	D25113	97.12	5.92	6.39
114	D25114	98.35	5.87	6.37
115	D25115	97.84	6.05	6.44
116	D25116	98.5	5.95	6.39
117	D25117	98.33	6.01	6.42
118	D25118	98.98	5.99	6.04
119	D25119	97.42	5.97	6.51
120	D25120	96.38	6.23	6.35
121	D25121	98.18	6.08	6.55
122	D25122	98.14	5.92	6.47
123	D25123	97.58	6.02	6.31
124	D25124	97.22	5.8	6.39
125	D25125	98.01	5.99	6.32
126	D25126	97.33	5.94	6.28
127	D25127	97.92	6.08	6.34
128	D25128	98.26	6.01	6.45
129	D25129	98.69	5.97	6.26
130	D25130	98.56	6.01	6.36
131	D25131	97.23	6.04	6.25
132	D25132	98.05	6.14	6.55
133	D25133	97.86	6.09	6.31
134	D25134	97.14	5.87	6.27
135	D25135	97.35	5.95	6.36
136	D25136	97.18	5.82	6.24
137	D25137	97.25	5.78	6.16
138	D25138	97.58	5.73	6.13
139	D25139	97.2	5.86	6.22
140	D25140	97.5	5.9	6.28
141	D25141	98.67	5.83	6.38
142	D25142	97.35	5.95	6.18
143	D25143	97.73	5.9	6.25
144	D25144	97.56	5.89	6.12
145	D25145	98.71	5.84	6.27
146	D25146	97.44	5.88	6.15
147	D25147	96.57	5.68	6.14
148	D25148	97.84	5.71	6.08
149	D25149	98.48	5.63	5.92
150	D25150	98.09	5.69	5.96
151	D25151	97.95	5.58	5.86
152	D25152	98.01	5.65	6.18
153	D25153	97.69	5.93	6.25

154	D25154	97.37	5.66	6.02
155	D25155	97.56	5.68	6.12
156	D25156	97.65	5.62	6.09
157	D25157	97.23	5.71	6.27
158	D25158	97.76	5.78	6.3
159	D25159	97.75	5.73	6.28
160	D25160	97.86	5.75	6.23
161	D25161	98.66	5.85	6.36
162	D25162	97.99	5.84	6.34
163	D25163	97.48	5.89	6.39
164	D25164	97.44	5.87	6.09
165	D25165	97.88	5.84	6.16
166	D25166	97.37	6.07	6.23
167	D25167	97.06	5.46	5.75
168	D25168	98.75	5.5	5.82
169	D25169	98.67	5.68	5.97
170	D25170	98.56	5.79	6.09
171	D25171	97.76	5.68	6.04
172	D25172	97.61	5.52	5.87
173	D25173	97.82	5.63	5.87
174	D25174	97.35	5.69	5.92
175	D25175	98.18	5.72	6.02
176	D25176	97.08	5.55	5.99
177	D25177	98.11	5.77	6.04
178	D25178	97.22	5.7	5.95
179	D25179	97.03	5.61	5.98
180	D25180	97.14	5.75	5.89
181	D25181	97.73	5.61	5.9
182	D25182	98.33	5.93	6.25
183	D25183	98.37	5.95	6.3
184	D25184	98.14	6.07	6.28
185	D25185	98.09	6.15	6.33
186	D25186	97.27	6.17	6.3
187	D25187	97.54	6.11	6.27
188	D25188	97.37	6.01	6.31
189	D25189	96.68	5.82	6.28
190	D25190	96.74	5.98	6.3
191	D25191	98.2	6.03	6.33
192	D25192	98.26	5.85	6.26
193	D25193	97.54	5.71	6.08
194	D25194	97.31	5.7	6.1
195	D25195	98.83	5.57	6.07
196	D25196	97.06	5.81	6.18
197	D25197	97.42	5.8	6.17
198	D25198	98.16	5.87	6.23
199	D25199	97.86	6.05	6.33

200	D25200	97.65	5.83	6.25
201	D25201	97.95	5.57	5.86
202	D25202	97.67	5.55	5.82
203	D25203	98.58	5.54	6.21
204	D25204	98.79	5.57	6.17
205	D25205	97.88	5.87	6.14
206	D25206	98.03	5.61	5.82
207	D25207	97.97	5.63	5.85
208	D25208	97.71	5.51	5.9
209	D25209	96.95	5.55	5.87
210	D25210	98.33	5.7	6.1
211	D25211	97.69	5.62	6.05
212	D25212	97.37	5.79	6.2
213	D25213	98.86	5.78	6.08
214	D25214	97.82	5.83	6.11
215	D25215	98.03	5.74	6.02
216	D25216	97.9	5.81	6.14
217	D25217	98.14	5.86	6.18
218	D25218	97.58	5.95	6.24
219	D25219	97.65	5.77	6.06
220	D25220	97.82	5.84	6.16
221	D25221	97.73	5.82	6.19
222	D25222	98.43	6.1	6.4
223	D25223	97.88	6.09	6.38
224	D25224	98.39	6.13	6.41
225	D25225	97.84	6.01	6.35
226	D25226	98.47	5.55	6.22
227	D25227	97.86	5.67	6.25
228	D25228	98.09	5.91	6.2
229	D25229	98.03	5.97	6.33
230	D25230	97.92	5.69	6.15
231	D25231	98.41	5.63	5.95
232	D25232	97.44	5.54	5.91
233	D25233	97.48	5.61	5.93
234	D25234	98.05	5.58	5.87
235	D25235	98.33	5.6	5.89
236	D25236	98.83	5.42	5.78
237	D25237	98.88	5.58	6.03
238	D25238	98.64	5.82	6.24
239	D25239	98.64	5.61	5.94
240	D25240	98.54	5.65	6.01
241	D25241	97.86	5.78	6.11
242	D25242	98.54	6.24	6.4
243	D25243	98.03	6.32	6.41
244	D25244	98.41	5.81	5.98
245	D25245	98.09	5.82	5.89

246	D25246	98.6	6	6.11
247	D25247	98.96	5.64	5.83
248	D25248	98.88	5.62	6.01
249	D25249	98.26	5.87	6
250	D25250	98.88	5.81	6.05
251	D25251	99.15	5.76	5.98
252	D25252	99.62	5.65	5.9
253	D25253	98.43	5.6	6.22
254	D25254	97.06	5.8	6.17
255	D25255	97.73	5.71	6.1
256	D25256	97.92	5.73	6.14
257	D25257	97.99	6.23	6.86
258	D25258	97.31	5.41	6.26
259	D25259	98.39	5.48	6.18
260	D25260	97.59	5.47	6.03
261	D25261	98.41	5.41	6.01
262	D25262	98.3	5.58	6.21
263	D25263	98.14	5.74	6.28
264	D25264	98.56	5.8	6.21
265	D25265	98.5	5.82	6.26
266	D25266	98.6	5.9	6.19
267	D25267	98.45	5.87	6.19
268	D25268	98.31	6	6.3
269	D25269	98.58	5.79	6.29
270	D25270	98.03	6.02	6.31
271	D25271	97.48	5.87	6.21
272	D25272	97.22	5.79	6.24
273	D25273	97.88	5.81	6.23
274	D25274	96.87	5.76	6.18
275	D25275	97.04	5.71	6.19
276	D25276	97.52	5.95	6.32
277	D25277	97.63	5.98	6.31
278	D25278	97.16	5.7	6.25
279	D25279	97.22	5.63	6.11
280	D25280	97.08	5.65	6.13
281	D25281	97.33	5.65	6.18
282	D25282	97.75	5.6	6.48
283	D25283	97.35	5.78	6.52
284	D25284	97.86	5.74	6.39
285	D25285	98.37	5.88	6.42
286	D25286	96.25	5.78	6.34

Chemical/Microbial Analysis for Finished product Result

S/N	BATCH NO	Dextrose Assay (95 – 105%)	pH (sample) (4.50 – 7.00)	HMF (NMT 0.25)	Sterility (MUST BE STERILE)
1	D25001	98.79	4.76	0.0700	STERILE
2	D25002	97.39	4.17	0.0710	STERILE
3	D25003	98.77	4.16	0.0760	STERILE
4	D25004	97.56	4.4	0.0730	STERILE
5	D25005	97.52	4.64	0.0770	STERILE
6	D25006	97.29	4.35	0.0800	STERILE
7	D25007	98.96	4.38	0.0740	STERILE
8	D25008	97.88	4.57	0.0760	STERILE
9	D25009	97.25	4.38	0.0750	STERILE
10	D25010	97.4	4.37	0.0710	STERILE
11	D25011	97.29	4.45	0.0780	STERILE
12	D25012	97.9	4.2	0.0720	STERILE
13	D25013	97.76	4.42	0.0750	STERILE
14	D25014	97.5	4.64	0.0770	STERILE
15	D25015	97.39	4.49	0.0730	STERILE
16	D25016	98.98	4.4	0.0700	STERILE
17	D25017	98.7	4.26	0.0790	STERILE
18	D25018	98.07	4.21	0.0690	STERILE
19	D25019	97.92	4.14	0.0740	STERILE
20	D25020	96.55	4.12	0.0800	STERILE
21	D25021	98.79	4.1	0.0700	STERILE
22	D25022	97.39	4.32	0.0710	STERILE
23	D25023	98.77	3.98	0.0760	STERILE
24	D25024	97.56	4.22	0.0730	STERILE
25	D25025	97.52	4.51	0.0770	STERILE
26	D25026	97.29	4.67	0.0800	STERILE
27	D25027	98.96	4.24	0.0740	STERILE
28	D25028	97.88	4.61	0.0760	STERILE
29	D25029	97.25	4.63	0.0750	STERILE
30	D25030	97.4	4.66	0.0710	STERILE
31	D25031	97.29	4.71	0.0780	STERILE
32	D25032	97.9	4.68	0.0720	STERILE
33	D25033	97.76	4.7	0.0750	STERILE
34	D25034	97.5	4.64	0.0770	STERILE
35	D25035	97.39	4.57	0.0730	STERILE
36	D25036	98.98	4.26	0.0700	STERILE
37	D25037	98.7	4.16	0.0790	STERILE
38	D25038	98.07	4.26	0.0690	STERILE
39	D25039	97.92	4.4	0.0740	STERILE
40	D25040	96.55	4.37	0.0800	STERILE
41	D25041	98.79	4.33	0.0700	STERILE

42	D25042	97.39	4.46	0.0710	STERILE
43	D25043	98.77	4.38	0.0760	STERILE
44	D25044	97.56	4.47	0.0730	STERILE
45	D25045	97.52	4.44	0.0770	STERILE
46	D25046	97.29	4.46	0.0800	STERILE
47	D25047	98.96	4.72	0.0740	STERILE
48	D25048	97.88	4.59	0.0760	STERILE
49	D25049	97.25	4.55	0.0750	STERILE
50	D25050	97.4	4.59	0.0710	STERILE
51	D25051	97.29	4.54	0.0780	STERILE
52	D25052	97.9	4.77	0.0720	STERILE
53	D25053	97.76	4.85	0.0750	STERILE
54	D25054	97.5	4.77	0.0770	STERILE
55	D25055	97.39	4.6	0.0730	STERILE
56	D25056	98.98	4.59	0.0700	STERILE
57	D25057	98.7	4.46	0.0790	STERILE
58	D25058	98.07	4.54	0.0690	STERILE
59	D25059	97.92	4.46	0.0740	STERILE
60	D25060	96.55	4.56	0.0800	STERILE
61	D25061	98.79	4.47	0.0700	STERILE
62	D25062	97.39	4.44	0.0710	STERILE
63	D25063	98.77	4.56	0.0760	STERILE
64	D25064	97.56	4.48	0.0730	STERILE
65	D25065	97.52	3.81	0.0770	STERILE
66	D25066	97.29	4.31	0.0800	STERILE
67	D25067	98.96	4.31	0.0740	STERILE
68	D25068	97.88	4.23	0.0760	STERILE
69	D25069	97.25	4.38	0.0750	STERILE
70	D25070	97.4	4.39	0.0710	STERILE
71	D25071	97.29	4.28	0.0780	STERILE
72	D25072	97.9	4.19	0.0720	STERILE
73	D25073	97.76	4.36	0.0750	STERILE
74	D25074	97.5	4.01	0.0770	STERILE
75	D25075	97.39	4.35	0.0730	STERILE
76	D25076	98.98	4.09	0.0700	STERILE
77	D25077	98.7	4.23	0.0790	STERILE
78	D25078	98.07	4.48	0.0690	STERILE
79	D25079	97.92	4.36	0.0740	STERILE
80	D25080	96.55	4.71	0.0800	STERILE
81	D25081	98.79	4.75	0.0700	STERILE
82	D25082	97.39	4.16	0.0710	STERILE
83	D25083	98.77	4.23	0.0760	STERILE
84	D25084	97.56	4.25	0.0730	STERILE
85	D25085	97.52	4.14	0.0770	STERILE
86	D25086	97.29	4.27	0.0800	STERILE
87	D25087	98.96	4.67	0.0740	STERILE

88	D25088	97.88	4.5	0.0760	STERILE
89	D25089	97.25	4.66	0.0750	STERILE
90	D25090	97.4	4.74	0.0710	STERILE
91	D25091	97.29	4.6	0.0780	STERILE
92	D25092	97.9	4.63	0.0720	STERILE
93	D25093	97.76	4.63	0.0750	STERILE
94	D25094	97.5	4.59	0.0770	STERILE
95	D25095	97.39	4.52	0.0730	STERILE
96	D25096	98.98	4.26	0.0700	STERILE
97	D25097	98.7	4.36	0.0790	STERILE
98	D25098	98.07	4.32	0.0690	STERILE
99	D25099	97.92	4.18	0.0740	STERILE
100	D25100	96.55	4.29	0.0800	STERILE
101	D25101	98.79	4.34	0.0700	STERILE
102	D25102	97.39	4.23	0.0710	STERILE
103	D25103	98.77	4.33	0.0760	STERILE
104	D25104	97.56	4.35	0.0730	STERILE
105	D25105	97.52	4.23	0.0770	STERILE
106	D25106	97.29	4.29	0.0800	STERILE
107	D25107	98.96	4.13	0.0740	STERILE
108	D25108	97.88	4.27	0.0760	STERILE
109	D25109	97.25	4.29	0.0750	STERILE
110	D25110	97.4	4.84	0.0710	STERILE
111	D25111	97.29	4.67	0.0780	STERILE
112	D25112	97.9	4.29	0.0720	STERILE
113	D25113	97.76	4.53	0.0750	STERILE
114	D25114	97.5	4.36	0.0770	STERILE
115	D25115	97.39	4.3	0.0730	STERILE
116	D25116	98.98	4.3	0.0700	STERILE
117	D25117	98.7	3.97	0.0790	STERILE
118	D25118	98.07	4.24	0.0690	STERILE
119	D25119	97.92	4.54	0.0740	STERILE
120	D25120	96.55	4.57	0.0800	STERILE
121	D25121	98.79	4.49	0.0700	STERILE
122	D25122	97.39	4.39	0.0710	STERILE
123	D25123	98.77	4.61	0.0760	STERILE
124	D25124	97.56	4.37	0.0730	STERILE
125	D25125	97.52	4.63	0.0770	STERILE
126	D25126	97.29	4.35	0.0800	STERILE
127	D25127	98.96	4.52	0.0740	STERILE
128	D25128	97.88	4.45	0.0760	STERILE
129	D25129	97.25	4.81	0.0750	STERILE
130	D25130	97.4	3.71	0.0710	STERILE
131	D25131	97.29	4.08	0.0780	STERILE
132	D25132	97.9	4.18	0.0720	STERILE
133	D25133	97.76	4.23	0.0750	STERILE

134	D25134	97.5	4.07	0.0770	STERILE
135	D25135	97.39	4.11	0.0730	STERILE
136	D25136	98.98	4.19	0.0700	STERILE
137	D25137	98.7	4.23	0.0790	STERILE
138	D25138	98.07	4.25	0.0690	STERILE
139	D25139	97.92	4.23	0.0740	STERILE
140	D25140	96.55	4.26	0.0800	STERILE
141	D25141	98.79	4.19	0.0700	STERILE
142	D25142	97.39	4.21	0.0710	STERILE
143	D25143	98.77	4.15	0.0760	STERILE
144	D25144	97.56	4.18	0.0730	STERILE
145	D25145	97.52	4.18	0.0770	STERILE
146	D25146	97.29	4.06	0.0800	STERILE
147	D25147	98.96	4.11	0.0740	STERILE
148	D25148	97.88	4.13	0.0760	STERILE
149	D25149	97.25	4.17	0.0750	STERILE
150	D25150	97.4	4.2	0.0710	STERILE
151	D25151	97.29	4.12	0.0780	STERILE
152	D25152	97.9	4.21	0.0720	STERILE
153	D25153	97.76	4.23	0.0750	STERILE
154	D25154	97.5	4.35	0.0770	STERILE
155	D25155	97.39	4.36	0.0730	STERILE
156	D25156	98.98	4.31	0.0700	STERILE
157	D25157	98.7	4.46	0.0790	STERILE
158	D25158	98.07	4.44	0.0690	STERILE
159	D25159	97.92	4.43	0.0740	STERILE
160	D25160	96.55	4.46	0.0800	STERILE
161	D25161	98.79	4.44	0.0700	STERILE
162	D25162	97.39	4.31	0.0710	STERILE
163	D25163	98.77	4.48	0.0760	STERILE
164	D25164	97.56	4.25	0.0730	STERILE
165	D25165	97.52	4.39	0.0770	STERILE
166	D25166	97.29	4.37	0.0800	STERILE
167	D25167	98.96	4.27	0.0740	STERILE
168	D25168	97.88	4.3	0.0760	STERILE
169	D25169	97.25	4.45	0.0750	STERILE
170	D25170	97.4	4.4	0.0710	STERILE
171	D25171	97.29	4.24	0.0780	STERILE
172	D25172	97.9	4.36	0.0720	STERILE
173	D25173	97.76	4.46	0.0750	STERILE
174	D25174	97.5	4.38	0.0770	STERILE
175	D25175	97.39	4.51	0.0730	STERILE
176	D25176	98.98	4.46	0.0700	STERILE
177	D25177	98.7	4.53	0.0790	STERILE
178	D25178	98.07	4.44	0.0690	STERILE
179	D25179	97.92	4.43	0.0740	STERILE

180	D25180	96.55	4.45	0.0800	STERILE
181	D25181	98.79	4.5	0.0700	STERILE
182	D25182	97.39	4.53	0.0710	STERILE
183	D25183	98.77	4.5	0.0760	STERILE
184	D25184	97.56	4.45	0.0730	STERILE
185	D25185	97.52	4.51	0.0770	STERILE
186	D25186	97.29	4.71	0.0800	STERILE
187	D25187	98.96	4.5	0.0740	STERILE
188	D25188	97.88	4.19	0.0760	STERILE
189	D25189	97.25	4.13	0.0750	STERILE
190	D25190	97.4	4.27	0.0710	STERILE
191	D25191	97.29	4.42	0.0780	STERILE
192	D25192	97.9	4.49	0.0720	STERILE
193	D25193	97.76	4.45	0.0750	STERILE
194	D25194	97.5	4.74	0.0770	STERILE
195	D25195	97.39	4.68	0.0730	STERILE
196	D25196	98.98	4.59	0.0700	STERILE
197	D25197	98.7	4.56	0.0790	STERILE
198	D25198	98.07	4.48	0.0690	STERILE
199	D25199	97.92	4.63	0.0740	STERILE
200	D25200	96.55	4.36	0.0800	STERILE
201	D25201	98.79	4.64	0.0700	STERILE
202	D25202	97.39	4.58	0.0710	STERILE
203	D25203	98.77	4.85	0.0760	STERILE
204	D25204	97.56	4.8	0.0730	STERILE
205	D25205	97.52	4.01	0.0770	STERILE
206	D25206	97.29	4.11	0.0800	STERILE
207	D25207	98.96	4.27	0.0740	STERILE
208	D25208	97.88	4.28	0.0760	STERILE
209	D25209	97.25	4.21	0.0750	STERILE
210	D25210	97.4	4.22	0.0710	STERILE
211	D25211	97.29	4.19	0.0780	STERILE
212	D25212	97.9	4.3	0.0720	STERILE
213	D25213	97.76	4.37	0.0750	STERILE
214	D25214	97.5	4.22	0.0770	STERILE
215	D25215	97.39	4.31	0.0730	STERILE
216	D25216	98.98	4.61	0.0700	STERILE
217	D25217	98.7	4.39	0.0790	STERILE
218	D25218	98.07	4.36	0.0690	STERILE
219	D25219	97.92	4.39	0.0740	STERILE
220	D25220	96.55	4.35	0.0800	STERILE
221	D25221	98.79	4.33	0.0700	STERILE
222	D25222	97.39	4.1	0.0710	STERILE
223	D25223	98.77	4.28	0.0760	STERILE
224	D25224	97.56	4.22	0.0730	STERILE
225	D25225	97.52	4.1	0.0770	STERILE

226	D25226	97.29	4.35	0.0800	STERILE
227	D25227	98.96	4.12	0.0740	STERILE
228	D25228	97.88	4.17	0.0760	STERILE
229	D25229	97.25	4.11	0.0750	STERILE
230	D25230	97.4	4.18	0.0710	STERILE
231	D25231	97.29	4.28	0.0780	STERILE
232	D25232	97.9	4.32	0.0720	STERILE
233	D25233	97.76	4.22	0.0750	STERILE
234	D25234	97.5	4.27	0.0770	STERILE
235	D25235	97.39	4.35	0.0730	STERILE
236	D25236	98.98	4.29	0.0700	STERILE
237	D25237	98.7	4.17	0.0790	STERILE
238	D25238	98.07	4.13	0.0690	STERILE
239	D25239	97.92	4.75	0.0740	STERILE
240	D25240	96.55	4.44	0.0800	STERILE
241	D25241	98.79	4.39	0.0700	STERILE
242	D25242	97.39	4.44	0.0710	STERILE
243	D25243	98.77	4.3	0.0760	STERILE
244	D25244	97.56	4.36	0.0730	STERILE
245	D25245	97.52	4.41	0.0770	STERILE
246	D25246	97.29	4.12	0.0800	STERILE
247	D25247	98.96	4.46	0.0740	STERILE
248	D25248	97.88	4.53	0.0760	STERILE
249	D25249	97.25	4.57	0.0750	STERILE
250	D25250	97.4	4.21	0.0710	STERILE
251	D25251	97.29	4.16	0.0780	STERILE
252	D25252	97.9	4.04	0.0720	STERILE
253	D25253	97.76	4.56	0.0750	STERILE
254	D25254	97.5	4.46	0.0770	STERILE
255	D25255	97.39	4.06	0.0730	STERILE
256	D25256	98.98	4.41	0.0700	STERILE
257	D25257	98.7	4.05	0.0790	STERILE
258	D25258	98.07	4	0.0690	STERILE
259	D25259	97.92	3.92	0.0740	STERILE
260	D25260	96.55	4.18	0.0800	STERILE
261	D25261	98.79	4.02	0.0700	STERILE
262	D25262	97.39	4.2	0.0710	STERILE
263	D25263	98.77	3.98	0.0760	STERILE
264	D25264	97.56	4.1	0.0730	STERILE
265	D25265	97.52	4.41	0.0770	STERILE
266	D25266	97.29	3.81	0.0800	STERILE
267	D25267	98.96	3.77	0.0740	STERILE
268	D25268	97.88	3.71	0.0760	STERILE
269	D25269	97.25	3.76	0.0750	STERILE
270	D25270	97.4	3.8	0.0710	STERILE
271	D25271	97.29	3.96	0.0780	STERILE

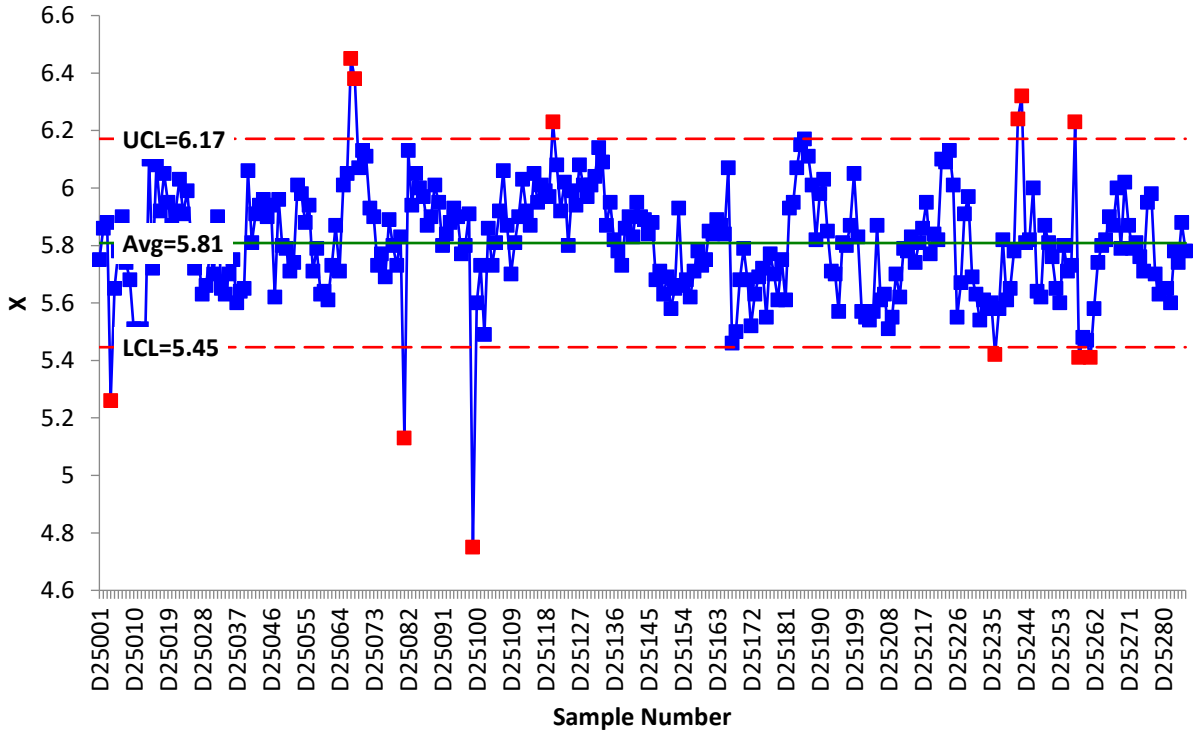
272	D25272	97.9	3.83	0.0720	STERILE
273	D25273	97.76	3.96	0.0750	STERILE
274	D25274	97.5	3.97	0.0770	STERILE
275	D25275	97.39	3.88	0.0730	STERILE
276	D25276	98.98	3.97	0.0700	STERILE
277	D25277	98.7	3.89	0.0790	STERILE
278	D25278	98.07	3.87	0.0690	STERILE
279	D25279	97.92	3.92	0.0740	STERILE
280	D25280	96.55	3.89	0.0800	STERILE
281	D25281	98.79	3.81	0.0700	STERILE
282	D25282	97.39	3.88	0.0710	STERILE
283	D25283	98.77	3.91	0.0760	STERILE
284	D25284	97.56	3.88	0.0730	STERILE
285	D25285	97.52	3.89	0.0770	STERILE
286	D25286	97.29	3.83	0.0800	STERILE

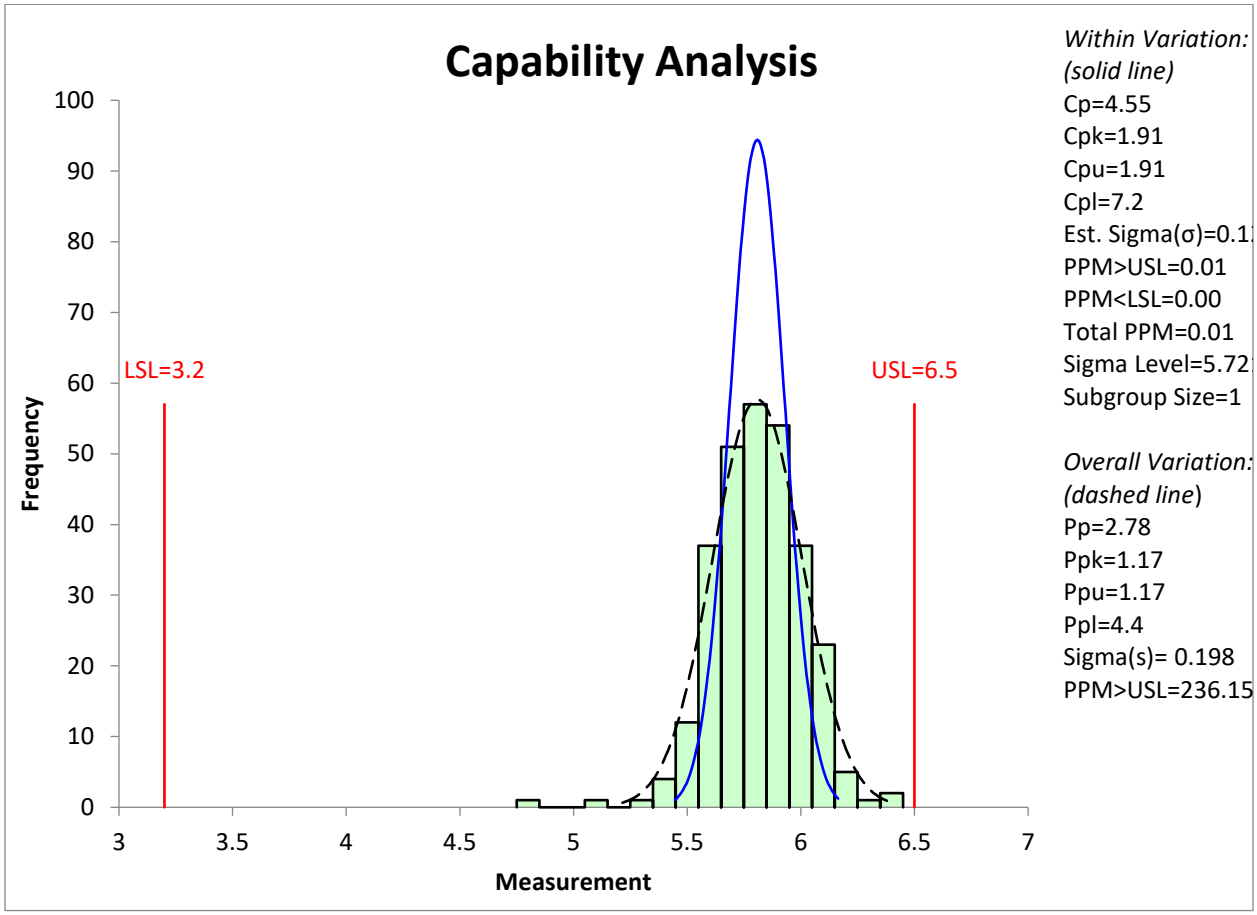
3.3.3 TREND ANALYSIS

Trend analysis of Dextrose content, pH, HMF, Pyrogen and Final yield %, were carried out on all the above batches.

Annexure I: pH trend for inprocess analysis of the product

Specification (3.2 -6.5**)**

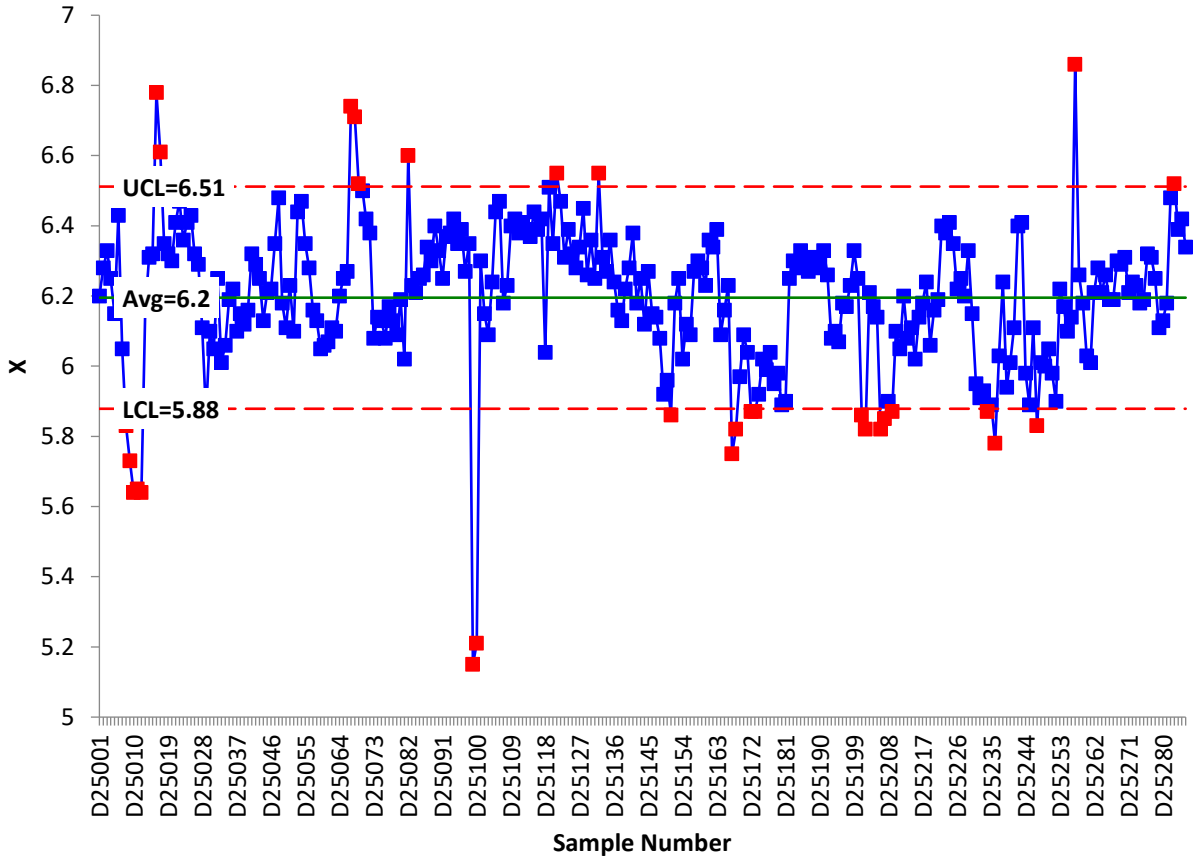


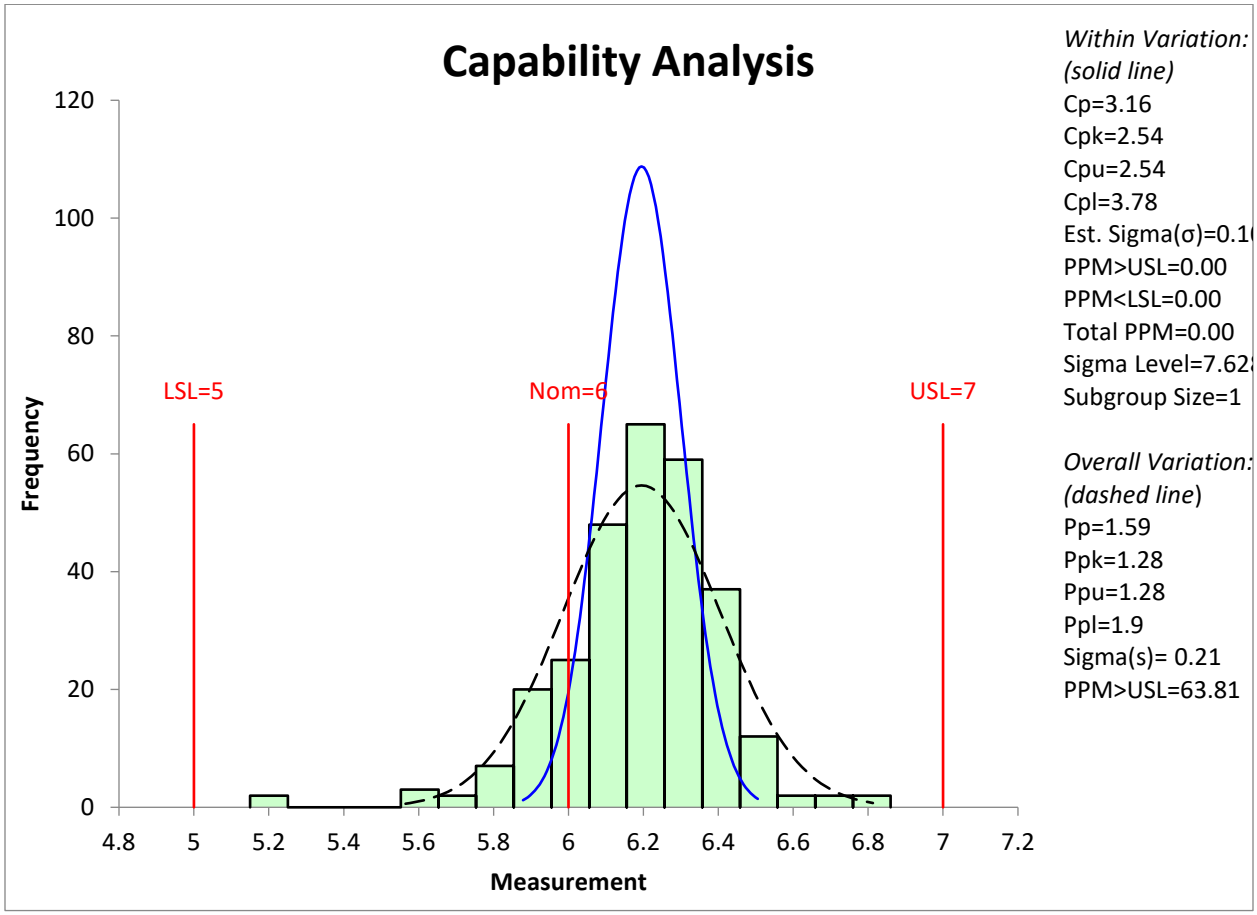


Mean	5.808601399
Standard deviation	0.19776861
Maximum	6.45
Minimum	4.75
Control limit: 5.22–6.49	
Alert limit: 5.41–6.20	
Minimum-Maximum: 4.75–6.45	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

Annexure II: pH trend for Water for Injection (WFI)

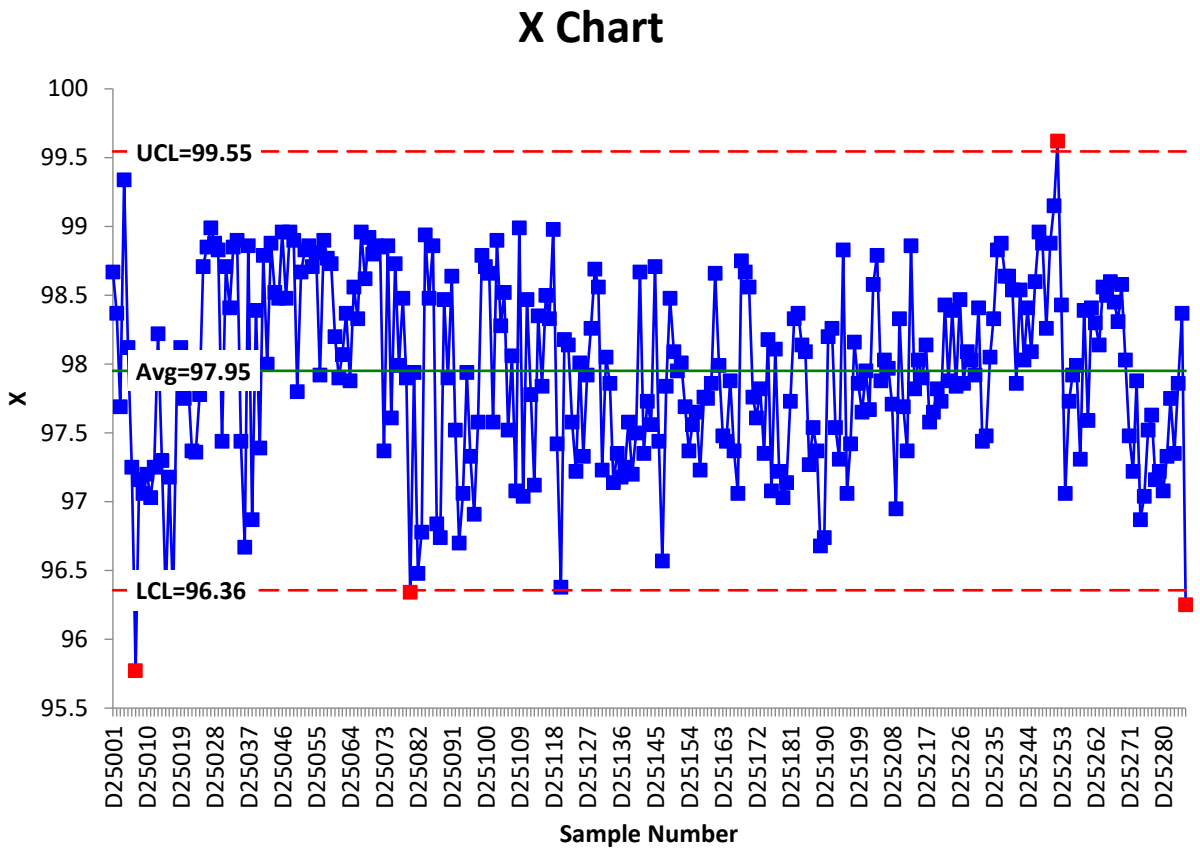
Specification (5.0 -7.0**)**

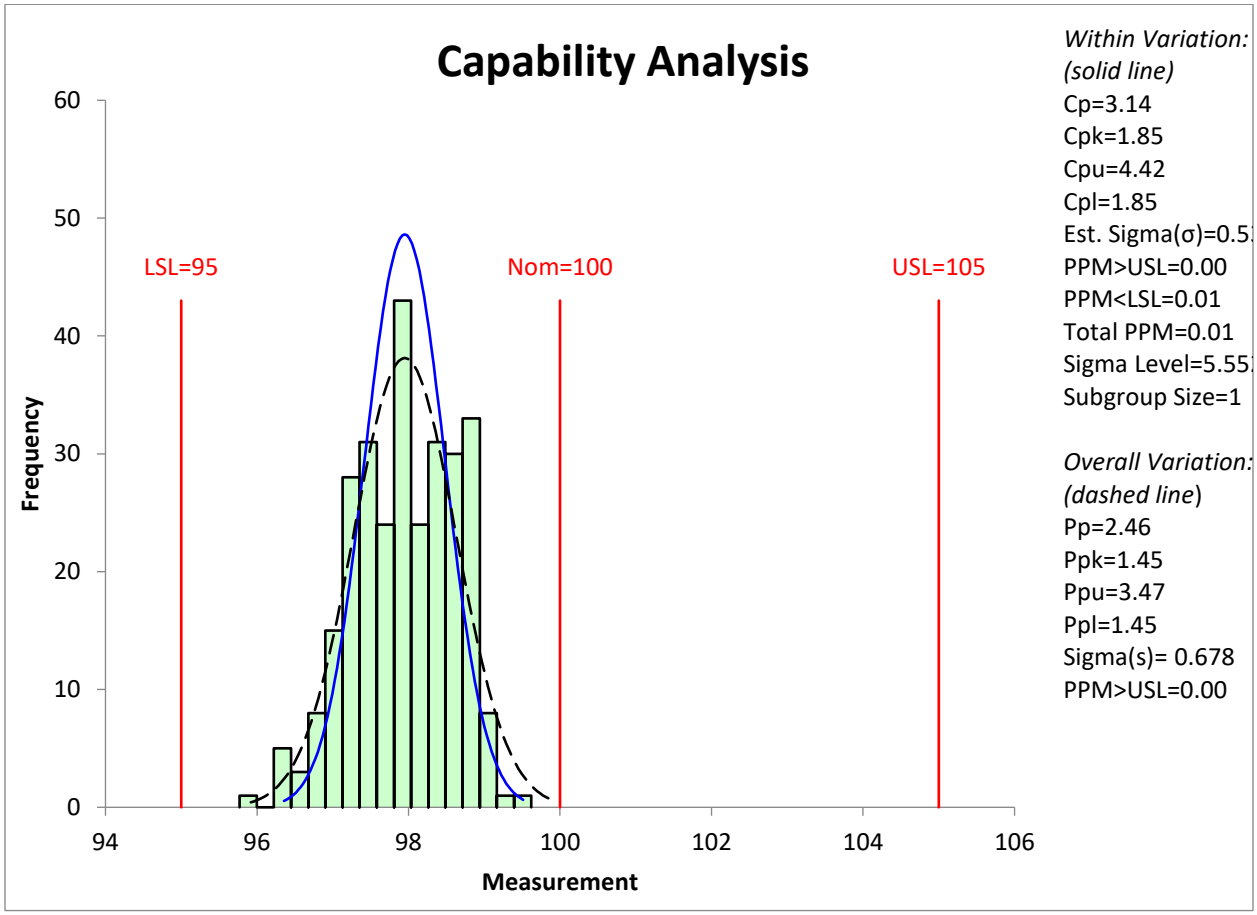




Mean	6.195175
Standard deviation	0.210082
Maximum	6.86
Minimum	5.15
Control limit: 5.56–6.83	
Alert limit: 5.78–6.62	
Minimum-Maximum: 5.15–6.86	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

Annexure III: Assay trend of in-process analysis of Dextrose in Dextrose 5% Injection
Specification of Dextrose (95 – 105%**)**

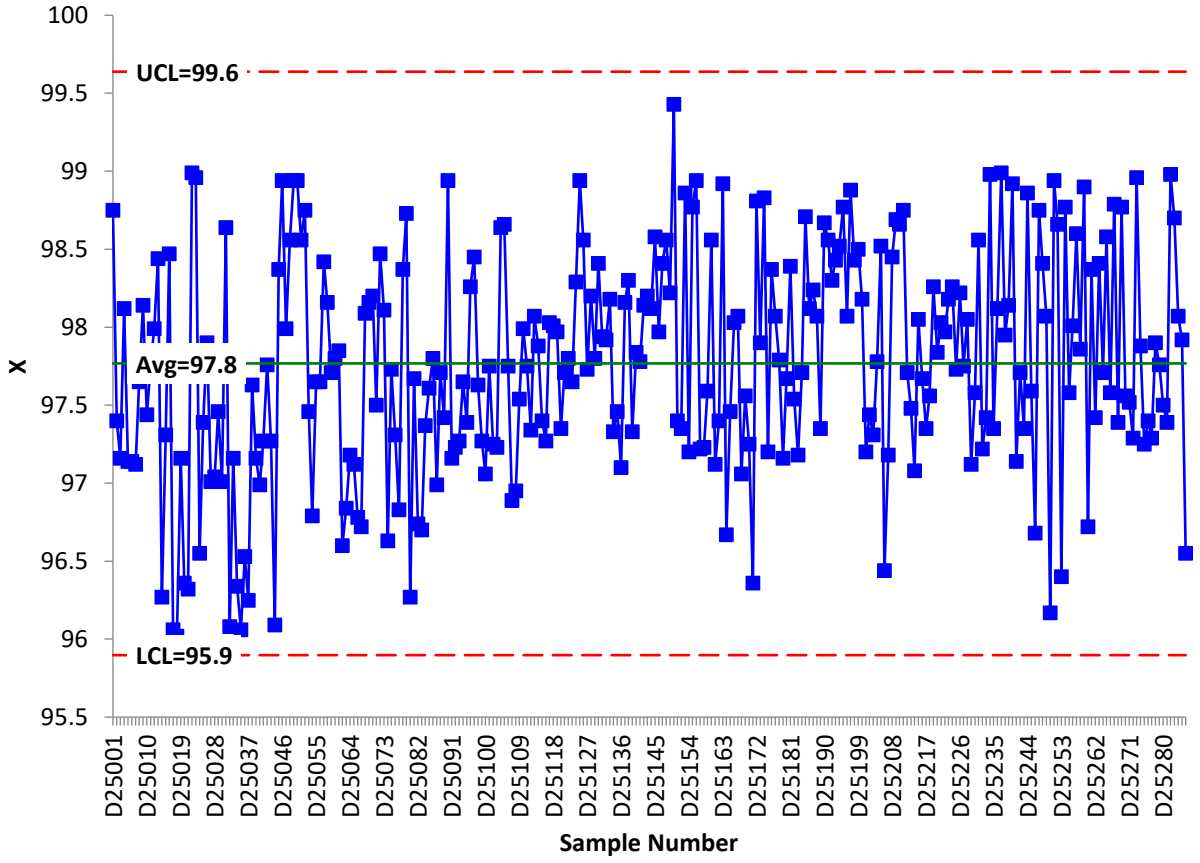


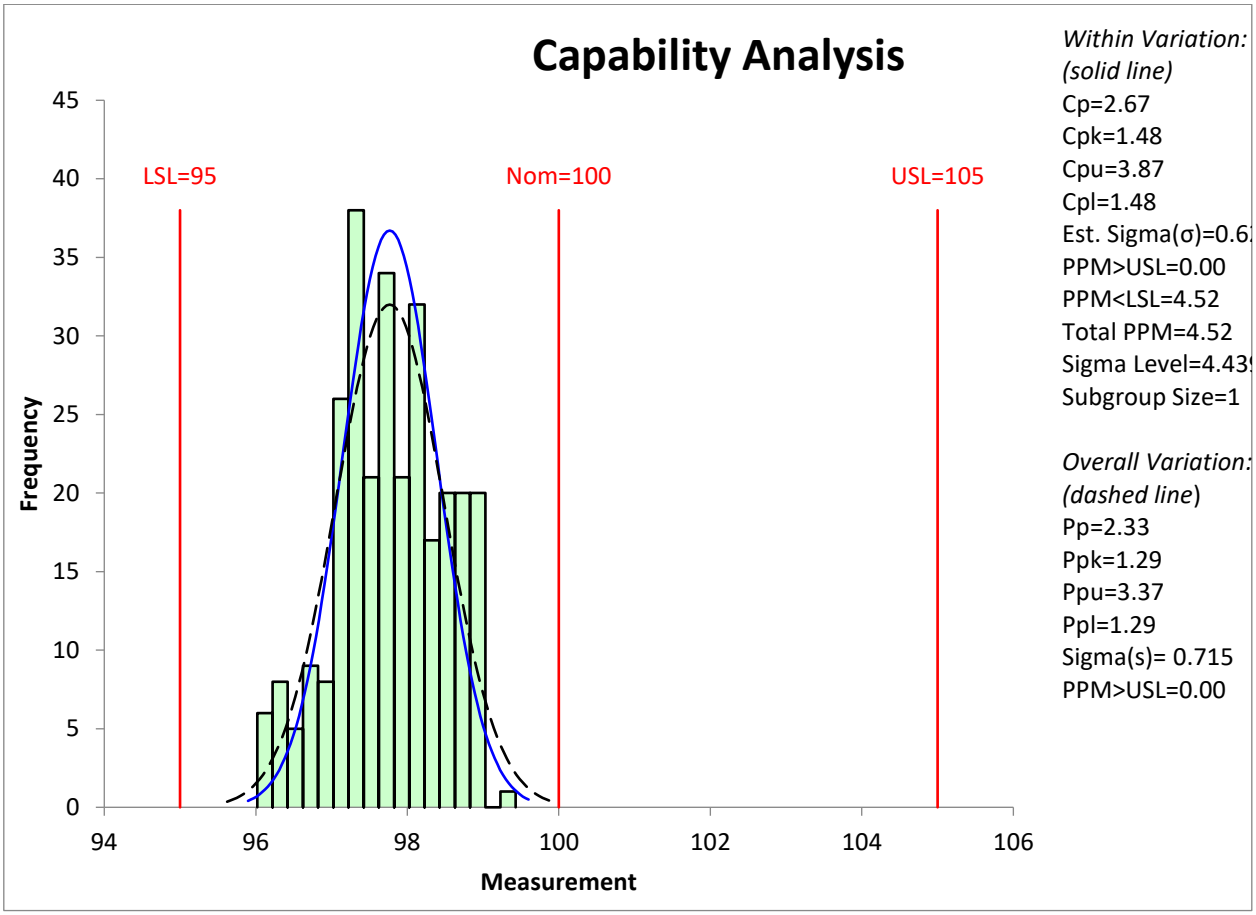


Mean	97.95062937
Standard deviation	0.67760747
Maximum	99.62
Minimum	95.77
Control limit: 95.92–99.98	
Alert limit: 96.60–99.31	
'Minimum-Maximum: 95.77–99.62	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

Annexure IV: Assay trend of finished product analysis of Dextrose in Dextrose 5% Injection

Specification of Dextrose (95 – 105%**)**

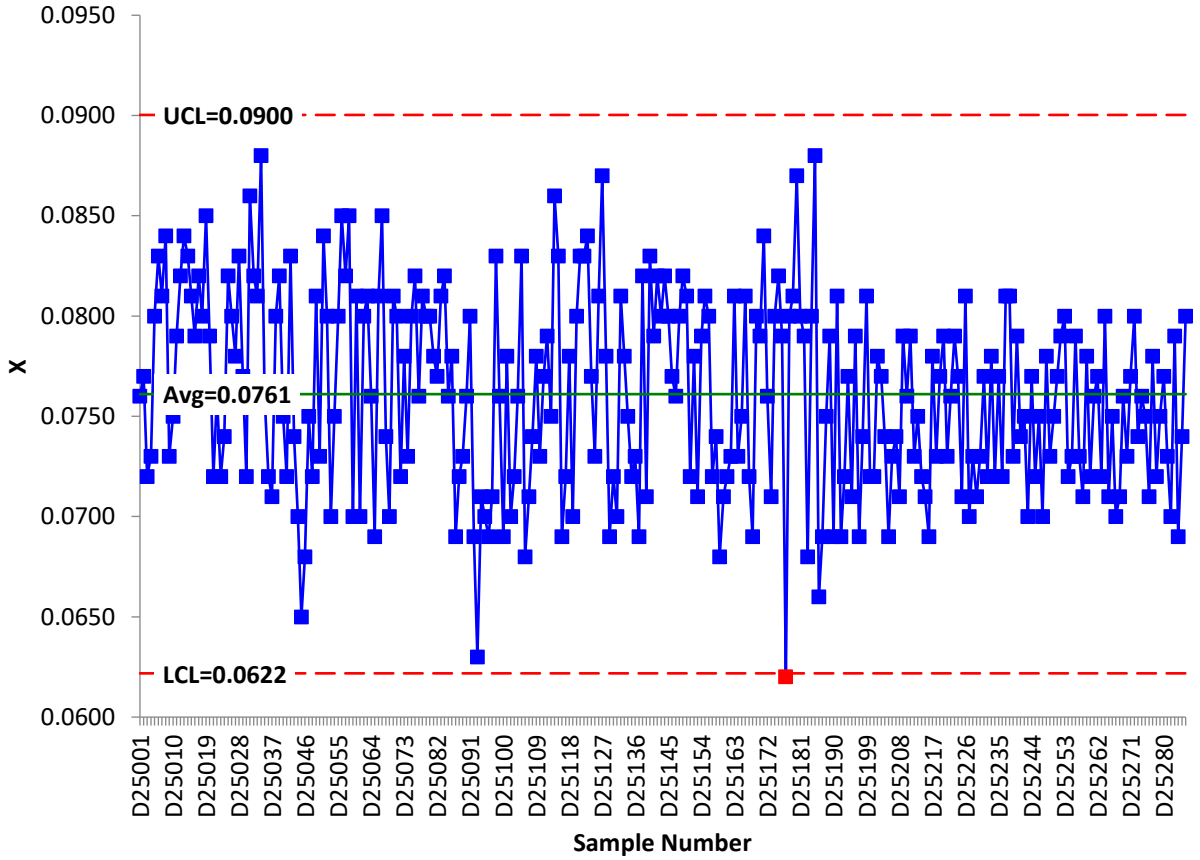


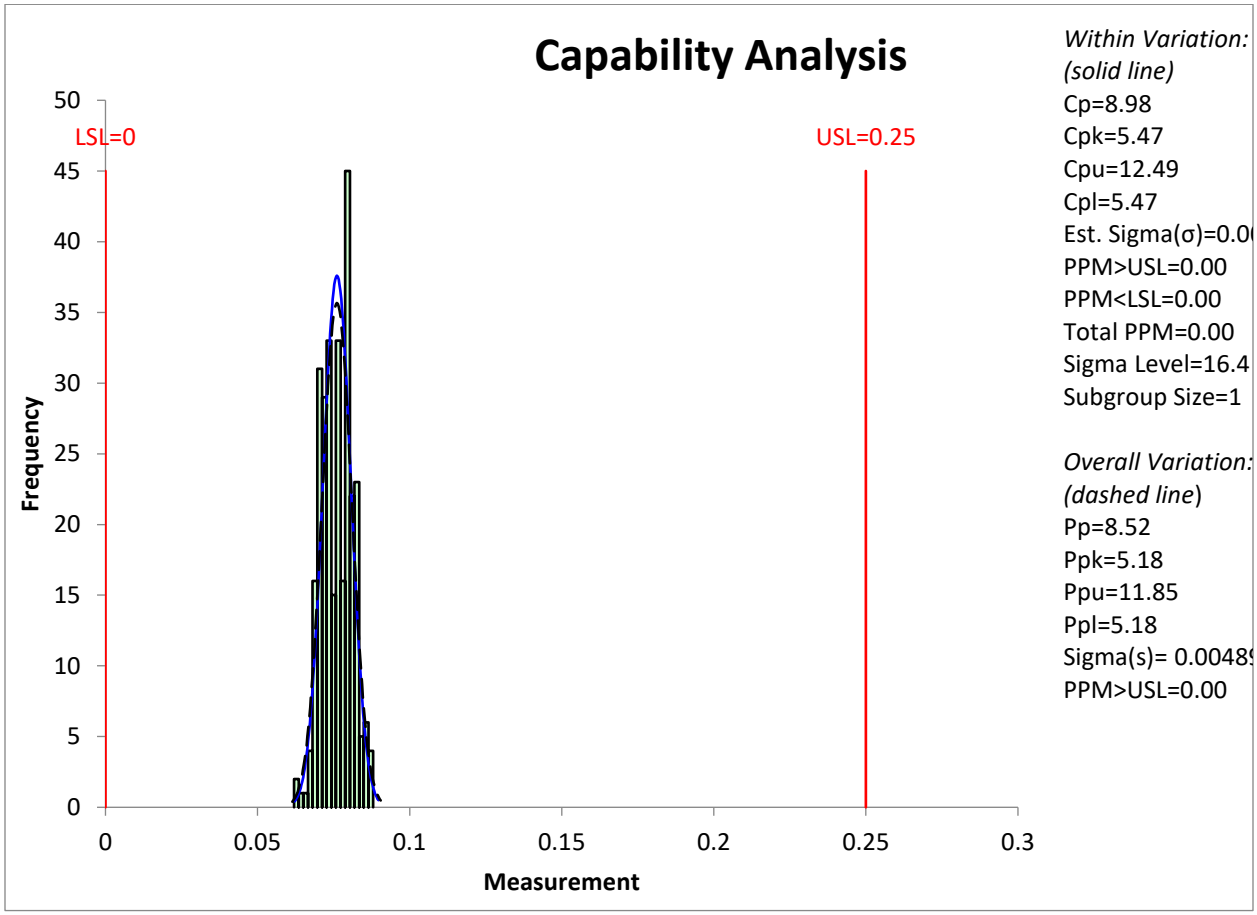


Mean	97.76765734
Standard deviation	0.715310583
Maximum	99.43
Minimum	96.02
Control limit: 95.62–99.91	
Alert limit: 96.34–99.20	
Minimum-Maximum: 96.02 –99.43	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

Annexure V: Assay trend of finished product analysis of HMF in Dextrose 50% Injection

Specification of Dextrose (95 – 105%)

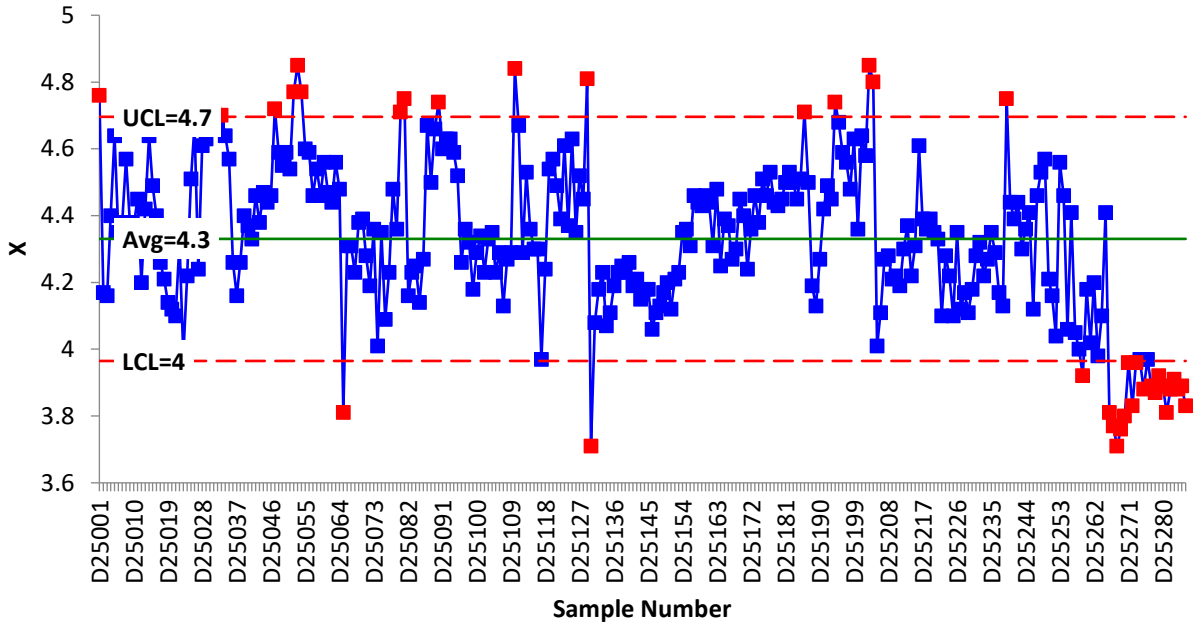


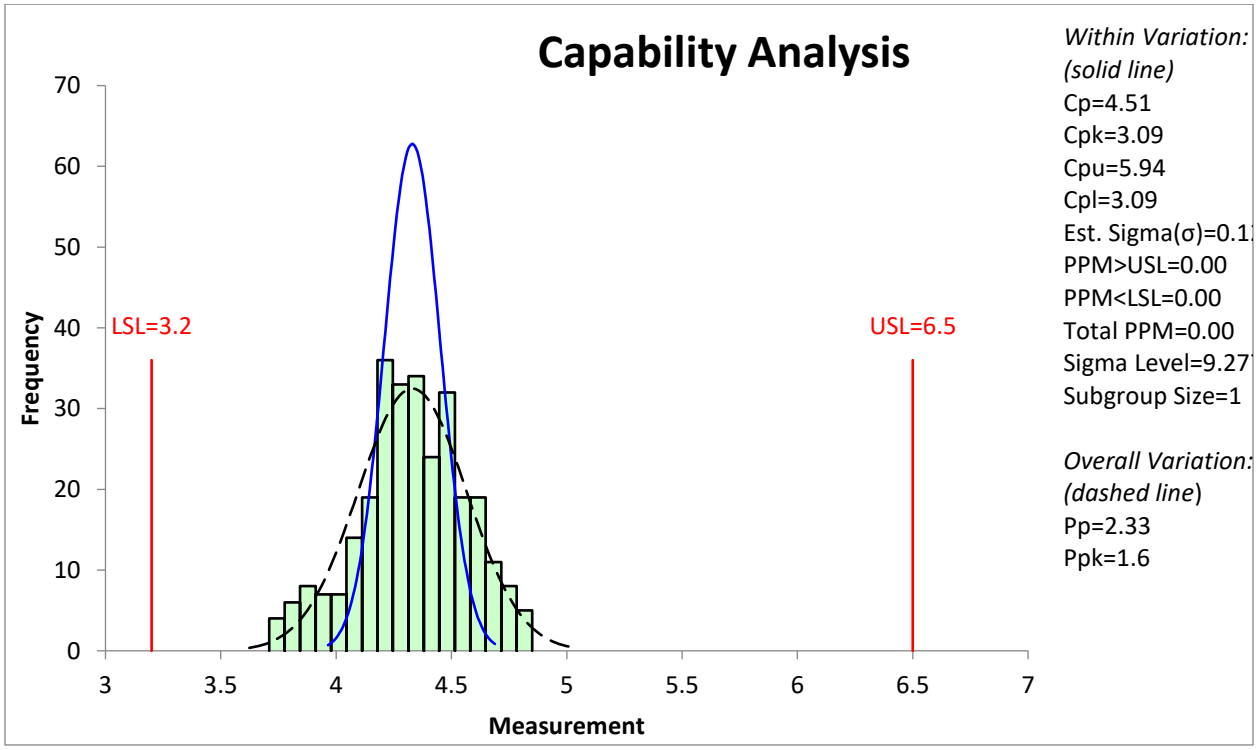


Mean	0.076104895
Standard deviation	0.004892835
Maximum	0.088
Minimum	0.062
Control limit: 0.061–0.091	
Alert limit: 0.066–0.086	
Minimum-Maximum: 0.062–0.088	
Alert limit within the specification limit, control limit existing within the specification limit.	

Annexure VI: pH trend of finished product analysis of Dextrose 5% Injection

Specification of pH (3.20-6.50**)**





Mean	4.33034965
Standard deviation	0.235553585
Maximum	4.85
Minimum	3.71
Control limit: 3.62–5.04	
Alert limit: 3.86–4.80	
Minimum-Maximum: 3.71–4.85	
Alert limit within the specification limit, control limit existing within the specification limit.	

SUMMARY OF PROCESS CAPABILITY

PROCESS CAPABILITY FOR INTERMEDIATE		
1	pH	1.91
2	Rinse Water	2.54
3	Dextrose	1.85
PROCESS CAPABILITY FOR PRODUCT		
1	pH	3.09
2	Dextrose	1.48
3	HMF	5.47

CQAs	Intermediate		Finished product	
	Control limit	Process capability	Process capability	Control limit
pH	CL within SL	1.91	3.09	CL within SL
Rinse water	CL within SL	2.54	-	-
Dextrose content	CL within SL	1.85	1.48	CL within SL
HMF	-	-	5.47	CL within SL

3.4 REVIEW OF BATCHES WITH OUT-OF-SPECIFICATION

No batch was recorded having out of specifications on any quality parameters.

3.5 REVIEW OF CRITICAL DEVIATIONS, NON-CONFORMANCES AND RELATED INVESTIGATIONS

3.5.1 PROCESS DEVIATIONS

No process deviation was reported from the standard processes for batches manufactured during the period Jan., 2025 to Dec., 2025.

3.5.2 YIELD DEVIATIONS

No observed deviation from the predetermined tolerance was observed.

3.5.3 NON-CONFORMANCE

No observed non-conformance from the predetermined tolerance was observed.

3.5.4 BATCH FAILURES

There was no batch failure observed during the period Jan., 2025 to Dec., 2025.

3.6 CHANGE CONTROL

There was no change done to the processes, equipment or analytical methods for the manufacture of Dextrose 50% Intravenous Injection during the year under review. As such, no change control was recorded throughout the period Jan., 2025 to Dec., 2025.

3.6.1 REGULATORY CHANGE

No regulatory changes as well as Master Formula Changes were made concerning these products throughout the manufactured batches till date.

3.6.2 CHANGES IN FORMULATION & PROCESS

No change is made in the formulation as well as process in regular practice. It has been observed that every batch manufactured, following consistent formulation and process, is giving consistent results which are proven by the trend analysis for all the parameters covered and expressed in graphical form.

3.7 STABILITY MONITORING PROGRAMME

The results of the real-time and accelerated stability monitoring program of all the commercial batches carried out showed consistency of all Critical Quality Attributes (CQAs). The product's Critical Quality Attributes (CQAs) like pH, Assay, Appearance, Pyrogenicity and Sterility are all found to be within the set specifications in conformity with the limits of the pharmacopeia. The stability report is attached for reference as BML/QA/SBP/001 (2025)

3.8 QUALITY-RELATED RETURNS, COMPLAINTS AND RECALLS

3.8.1 MARKET COMPLAINTS

Up till now there is no market complaint received for this product during the period Jan. 2025 to Dec. 2025.

3.8.2 RETURNED GOODS FROM MARKET

No finished goods were returned from the market for the period Jan., 2025 to Dec., 2025 due to any quality problem

3.8.3 BATCH RECALLS

Up till now there is no batch recall for product during the period Jan., 2025 to Dec., 2025.

3.8.4 SALVAGED GOODS

There is no salvaging of goods for the period of Jan., 2025 to Dec., 2025.

3.9 CORRECTIVE ACTIONS

No critical, major or minor deviations have been observed during the year under review. As such, no noteworthy corrective action was necessary. Few observed deviations during the year under review fall under the category of Others, which has to do with colour or print on tertiary packaging.

3.10 VALIDATED ANALYTICAL AND MANUFACTURING PROCEDURES

MANUFACTURING PROCEDURE

Equipment cleaning and sterilization
Dispensing of the raw material
Solution preparation
Batch coding of PVC bag (packaging material)
Solution filling
Sterilization of the filled solution in qualified autoclave
Final packaging of the sterilized solutions

ANALYTICAL PROCEDURE

In-process testing of the cleanliness of equipment
In-process chemical testing of the semi-finished products for critical quality attributes like pH, assay
Chemical testing of the finished products
Microbiological tests on the finished product-
Sterility tests
Bacterial endotoxin tests (Pyrogen and Limulus Amoebocyte Lysate)

SUMMARY FOR PROCESS VALIDATION:

We have manufactured 8 batches of Dextrose 50% Intravenous Injection 100ml with same manufacturing formula from Jan., 2025 till Dec. 2025 and yearly APQR data shows satisfactory trend throughout the

period. Also, the retained samples prove that the product manufactured with the existing formulation is stable.

ENVIRONMENTAL CONDITIONS DURING MANUFACTURING OPERATION

The data of environmental conditions during the manufacturing of all batches were reviewed and recorded in the APQR. The environmental conditions of the critical areas were properly monitored in the raw material store, mixing unit, filling unit, batch coding unit and storage areas.

Environmental Conditions
Std Temp. : NMT 30°C
Relative Humidity: NMT 70%

CRITICAL EQUIPMENT PERFORMANCE

The critical equipment used during the manufacturing and analysis of the product were identified and overall performance of these equipment were evaluated by reviewing the breakdown and preventative maintenance records.

3.11 REVIEW OF OBSERVED ADVERSE DRUG REACTIONS

No adverse drug reaction (ADR) was observed with the product during the year under review.

3.12 REVIEW OF PERIODIC RISK-BENEFIT EVALUATION REPORTS

Bioflex Dextrose 5% Intravenous Injection is an intravenous carbohydrate replenishment preparation officially approved for sale in Nigeria since the year 2020. The periodic safety update reports submitted for the product covering January-December 2025 indicated a favourable risk-benefit profile, with no adverse risk signals identified during the review period. All columns under the spontaneous safety signal and risk report evaluation for the System Organ Class (SOC) returned zero entries. About 870,000 bags of Bioflex Dextrose 5% Intravenous Injection were manufactured and distributed across the nation in the year under review, and no report indicating either serious or non-serious early signal detection and/or safety risks on any essential System Organ Class was recorded.

3.13 REVIEW OF POST MARKETING SURVEILLANCE

The reports of clinical and therapeutic performance of the products from major health institutions across the nation have proven the product to be safe and efficacious. No adverse reaction has been reported, no report of product contamination nor pyrogenicity has been reported with the product within the year under review.

3.14 COMPARATIVE PRODUCT QUALITY REVIEW
CONCLUSION

The data indicates that all Critical Quality Attributes (CQAs) evaluated for both intermediate and finished product stages are within established control and specification limits, demonstrating that the manufacturing process is generally well controlled. Process capability values for pH (1.91 at intermediate and 3.09 at finished product) show strong performance, with the finished product stage exhibiting excellent consistency and reduced variability. Rinse water also demonstrates high process capability at the intermediate stage (2.54), confirming stable control of this parameter within the production process.

For dextrose content, the process remains capable at both stages, with values of 1.85 (intermediate) and 1.48 (finished product), although a slight reduction at the finished stage suggests marginally increased variability after final processing while remaining within acceptable limits. HMF, assessed at the finished product stage only, shows an exceptionally high capability value of 5.47, indicating very tight control of impurity formation. Overall, the results confirm a robust and consistent manufacturing process with all CQAs performing within acceptable quality standards.